



Cerebrovascular Accident (Stroke) Risk Analysis

Chronic-Clinic Triage Screening Model: Regional Referral Hospital, Uganda

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AI Model Deployment and Scalability: Assignment 2 (September 2026)

Context & Task Definition

- High-volume Regional Referral Hospital in Uganda: demographic, clinical & lifestyle data collected on patients attending the hospital
- Goal: flag high-risk patients for fast-track into the chronic clinic, per the Ministry of Health's June 2025 service-integration guidelines
- Known_Patients_01 (labelled, historical outcomes) trains and evaluates the models
- New_Patients_01 (no recorded outcome) is the operational batch the hospital needs risk scores for
- Nature of the task: stroke is binary (1/0) → binary classification, not regression
- The deliverable is a risk score (predicted probability), not just a hard label, so ROC-AUC (ranking quality) is the primary metric, alongside recall and precision

~5%

stroke prevalence in Known_Patients_01

Classification

not regression – predicted risk score
used for triage

MoH June 2025

service-integration guideline for the
chronic clinic pathway



Dataset Description

4,644

Known_Patients_01
rows

466

New_Patients_01 rows

12

attributes per patient

4.0%

missing bmi values

id	Unique patient identifier
gender	Male / Female / Other
age	Age of the patient
hypertension	0 / 1 (has hypertension)
heart_disease	0 / 1 (has heart disease)
ever_married	No / Yes

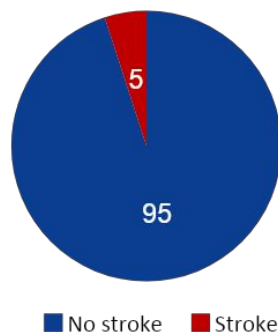
work_type	Children / Govt job / Never worked / Private / Self-employed
Residence_type	Rural / Urban
avg_glucose_level	Average blood glucose level
bmi	Body mass index
smoking_status	Formerly smoked / Never smoked / Smokes / Unknown
stroke	1 = stroke occurred, 0 = did not (target)

Target: predict whether a patient in New_Patients_01 is high-risk for stroke, to prioritise chronic-clinic fast-track review.

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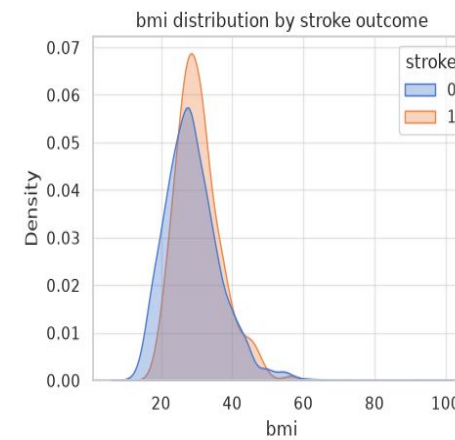
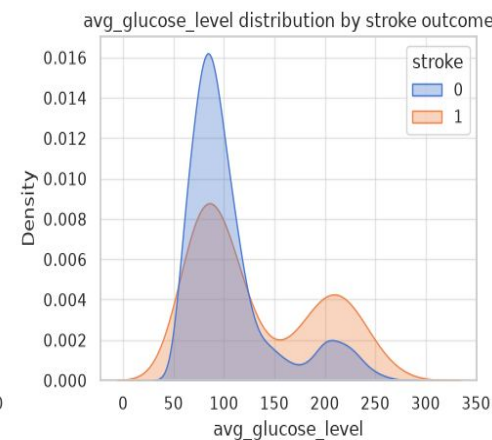
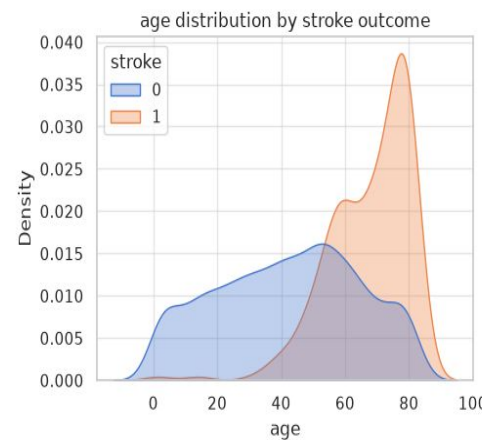
EDA – Class Imbalance & Key Distributions

Target class share (n=4,644)



232

stroke cases out of 4,644 patients (5.0%)

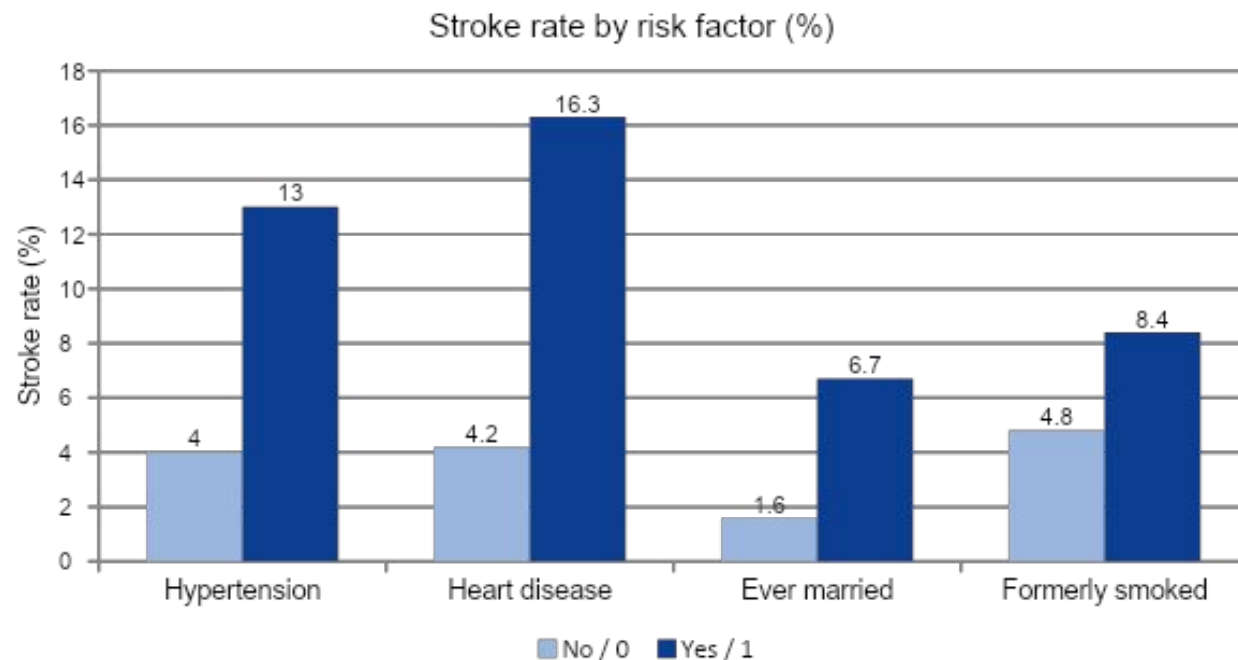


age / avg_glucose_level / bmi – kernel density by stroke outcome

- Stroke prevalence is ~5%, a strongly imbalanced classification problem; accuracy alone is a misleading headline metric
- Stroke patients skew markedly older, with a longer right tail on avg_glucose_level; bmi separates the two classes far less cleanly
- Missingness is confined to bmi (~4% of rows) with no structured pattern, consistent with missing-at-random, supporting simple median imputation

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EDA – Risk Factors & Correlation



Sampling note: single-hospital data, case-mix may not generalise to every clinic in Uganda.

Correlation with stroke

age	0.25
avg_glucose_level	0.13
hypertension	0.12
heart_disease	0.12
bmi	0.04

- No strong collinearity among numeric predictors ($|r| < 0.3$ pairwise), multicollinearity is not a major concern for Logistic Regression
- hypertension, heart_disease, ever_married and formerly-smoked status all associate with visibly higher stroke rates, consistent with established clinical risk factors

Data Preparation & Splitting

Variable type	Columns	Encoder used	Why
Continuous numeric	age, avg_glucose_level, bmi	Median impute + StandardScaler	Common scale for distance/gradient models; harmless for trees
Binary numeric	hypertension, heart_disease	Passed through as-is	Already 0/1, no encoding needed
Nominal categorical	gender, ever_married, work_type, Residence_type, smoking_status	One-Hot Encoding	No ordinal relationship between categories

- Bias/variance/noise: single-hospital sampling bias; ~5% class imbalance is a form of label/metric bias baked into accuracy itself; KNN/unpruned trees/SVM risk high variance on the minority class, mitigated with CV, regularisation & class weighting
- Linearity check (logit scale): age tracks close to a straight line (favours Logistic Regression); avg_glucose_level and bmi show a bend, favours tree-based / boosting models
- Stratified 80/20 split on Known_Patients_01 preserves the ~5% prevalence in both partitions (train 3,714 rows / test 929 rows); New_Patients_01 held out entirely as the unseen operational batch



Group A – Classical Classifiers

- Every model fit with `class_weight='balanced'`; tuned via 5-fold stratified CV (RandomizedSearchCV) optimising ROC-AUC – so Accuracy/F1 don't always rise together with tuning

Model	Accuracy (Before → After)	F1 Score (Before → After)	Hyperparameters Tuned
Logistic Regression	0.743 → 0.730	0.241 → 0.232	C = 0.01, penalty = l2
Decision Tree	0.767 → 0.760	0.229 → 0.254	max_depth = 3, min_samples_leaf = 1, criterion = gini
KNN	0.950 → 0.950	0.000 → 0.000	n_neighbors = 31, weights = uniform, p = 2
Naive Bayes	0.383 (not tuned)	0.133 (not tuned)	No hyperparameters searched (GaussianNB)
SVM	0.741 → 0.709	0.199 → 0.224	C = 0.1, gamma = 0.1, kernel = rbf
Random Forest	0.785 → 0.757	0.248 → 0.261	n_estimators=200, max_depth=None, min_samples_leaf=20, max_features=sqrt

Group B – Boosting & Imbalance Techniques

- `scale_pos_weight = 18.97` (majority:minority ratio) used for imbalance-aware XGBoost
- SMOTE oversampling (186 → 3,528 minority rows) lifts recall by rebalancing the training set itself, at some cost to precision
- Optuna hyperparameter search (tree depth, learning rate, regularisation) tunes XGBoost against CV ROC-AUC
- LightGBM with a custom focal-loss objective is competitive on ROC-AUC / log-loss – with no oversampling needed at all
- Plain XGBoost trails AdaBoost here: boosting power alone doesn't fix imbalance – it needs an explicit correction

Model	Accuracy (Before → After)	F1 Score (Before → After)	Hyperparameters Tuned
AdaBoost	0.950 → 0.950	0.000 → 0.000	<code>base_max_depth = 2</code> , <code>base_min_samples_leaf = 24</code> , <code>n_estimators = 140</code> , <code>learning_rate = 0.107</code>
XGBoost	0.892 → 0.755	0.275 → 0.240	<code>max_depth = 3</code> , <code>learning_rate = 0.026</code> , <code>n_estimators = 138</code> , <code>subsample = 0.677</code> , <code>colsample_bytree = 0.670</code> , <code>min_child_weight = 6</code> , <code>reg_alpha = 0.062</code> , <code>reg_lambda = 0.536</code>

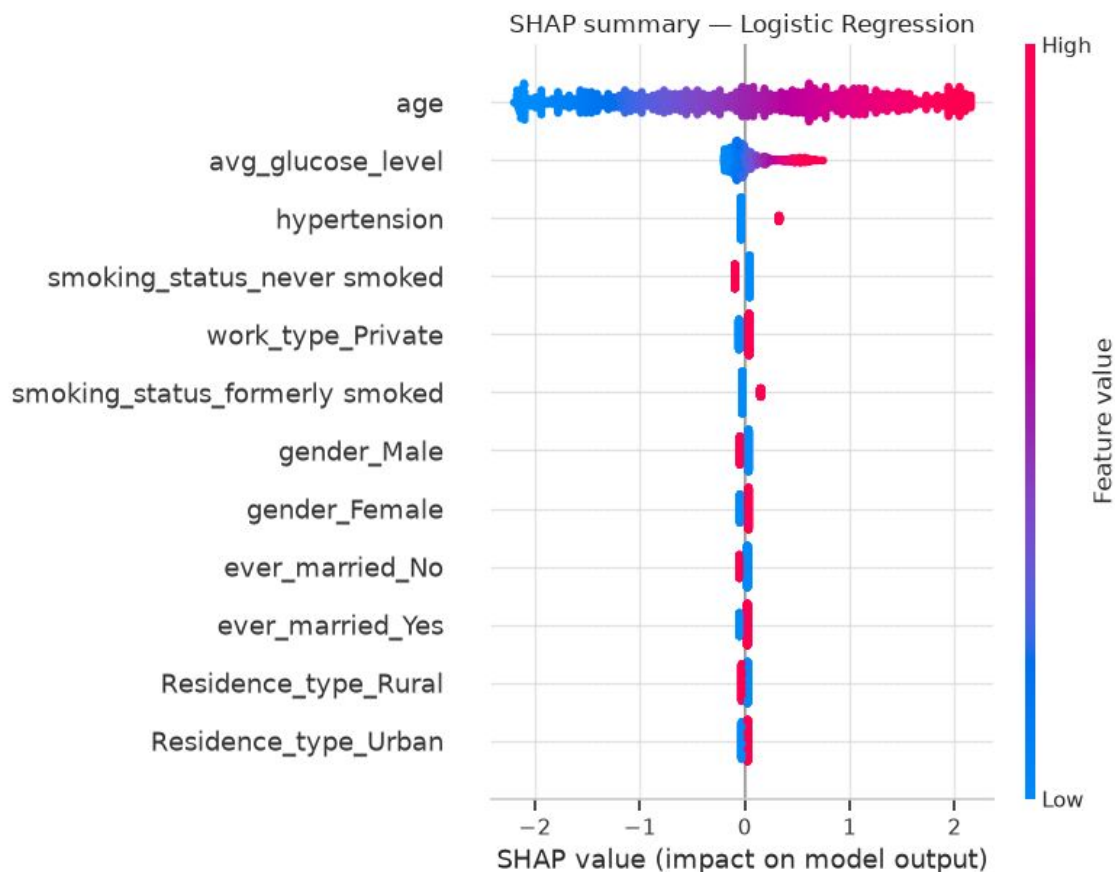
Model Comparison & Best Model Selection

- Best Group A: Logistic Regression (ROC-AUC 0.866)
- Best Group B (boosting family): XGBoost + Optuna tuned (ROC-AUC 0.859)
- **Best overall: Logistic Regression (ROC-AUC 0.866)**
- Recommended for operational triage scoring and as the drift-monitoring baseline

Group	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Log-Loss
A	Logistic Regression	0.730	0.135	0.826	0.232	0.866	0.494
B	AdaBoost + Optuna tuned	0.950	0.000	0.000	0.000	0.862	0.289
B	XGBoost + Optuna tuned	0.755	0.142	0.783	0.240	0.859	0.426
A	SVM	0.709	0.129	0.848	0.224	0.856	0.155
B	AdaBoost (baseline)	0.950	0.000	0.000	0.000	0.855	0.407
A	Random Forest	0.757	0.154	0.870	0.261	0.845	0.423
B	LightGBM + focal loss	0.950	0.000	0.000	0.000	0.839	0.190
A	Decision Tree	0.760	0.150	0.826	0.254	0.837	0.514
A	Naive Bayes	0.383	0.072	0.957	0.133	0.810	4.026
B	XGBoost (baseline)	0.892	0.207	0.413	0.275	0.797	0.263
A	KNN	0.950	0.000	0.000	0.000	0.763	0.407
B	XGBoost + SMOTE	0.925	0.167	0.130	0.146	0.732	0.234

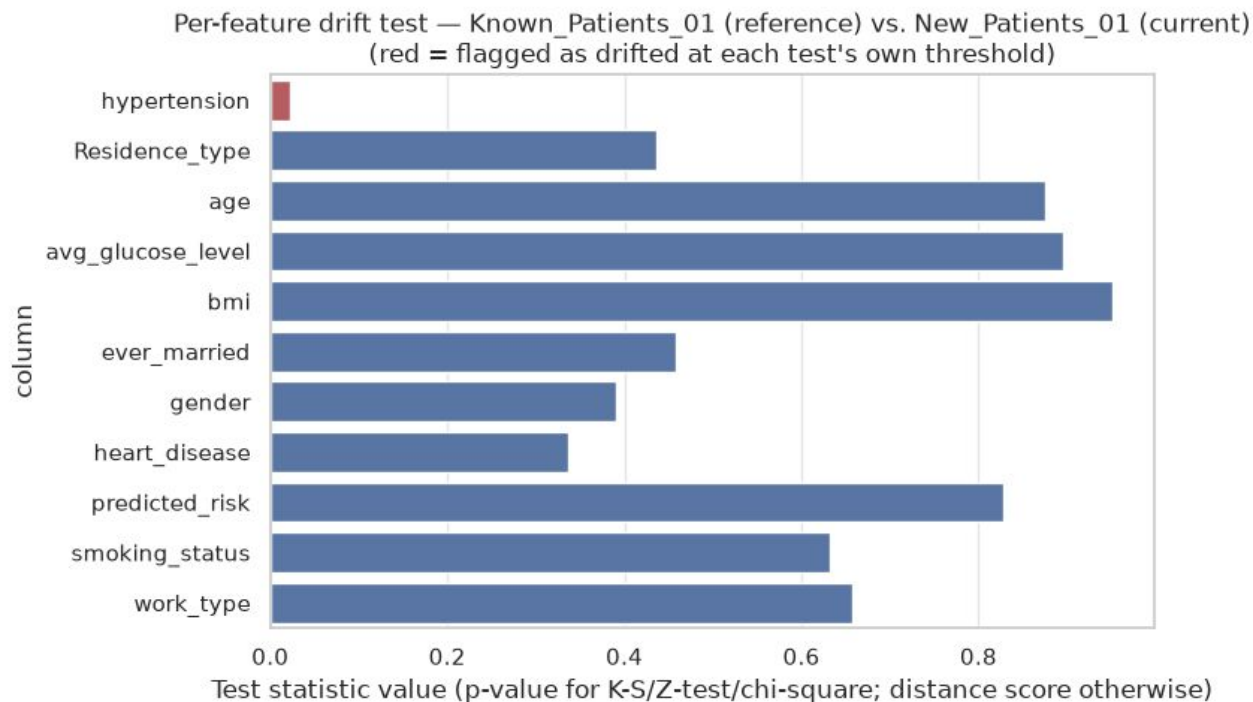
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Explainability – SHAP



- Explainer matched to model family: LinearExplainer for Logistic Regression (exact, fast); TreeExplainer for the tuned XGBoost (exact via tree structure)
- **age, avg_glucose_level and hypertension carry the largest average impact on predicted risk across both best models – matching the clinical risk factors already seen in EDA**
- bmi and work_type / smoking_status contribute more modestly
- Convergence between the linear and boosting models is reassuring evidence the models learn genuine clinical signal, not spurious correlation
- Counterfactual explanations (e.g. “what glucose change flips this patient's flag”) are a natural extension, left as future work

Drift Monitoring & Chronic-Clinic Triage



Evidently AI: per-column drift test, Known_Patients_01 (reference) vs. New_Patients_01 (current)

- 1 of 11 monitored columns drifted (hypertension, Z-test $p=0.023$), predicted-risk distribution itself is stable (K-S $p=0.83$)
- New_Patients_01 is statistically consistent with the training population, risk scores can be trusted at face value
- Triage threshold (Youden's J): 0.626, not the default 0.5, since `class_weight='balanced'` shifts calibration
- 466 patients scored → 89 flagged high-risk (19.1%) for chronic-clinic fast-track (mean predicted risk 0.347)



Conclusions & Recommendations

- Task nature: binary classification on a rare (~5%) outcome, accuracy is a poor headline metric; ROC-AUC, recall and precision drive the real modelling decisions
- Best models: Logistic Regression (Group A) and XGBoost + Optuna tuned (Group B, boosting family) gave the strongest, most balanced ROC-AUC once class-weighting / imbalance handling was applied
- Interpretability: SHAP on both best models converges on the same clinically-sensible top risk drivers (age, glucose, hypertension), supports clinician trust
- Drift: New_Patients_01 is broadly consistent with the training population; drift should be re-checked on every new intake batch
- **Recommendation: use the risk score to triage patients into the chronic clinic, per the MoH June 2025 guideline; set the operating threshold by desired recall, not a default 0.5**
- Limitations: single-hospital data may not generalise; bmi missingness/noise remains a residual uncertainty; periodic retraining and re-validation is recommended



References

- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16, 321-357.
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal Loss for Dense Object Detection. *ICCV 2017*.
- Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *NeurIPS*.
- Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. *KDD 2019*.
- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *KDD 2016*.
- Ke, G. et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *NeurIPS*.
- Evidently AI documentation: <https://docs.evidentlyai.com>
- Ministry of Health, Uganda: Service Integration Guidelines (June 2025), as referenced in the assignment brief.